Shari’s Week 0 Day 6 Vital Sign Report

Mean Arterial Pressure (MAP)

Mean arterial pressure (MAP) is the average pressure within the arteries throughout one cardiac cycle and serves as a key indicator of whether adequate blood flow is being maintained to vital organs such as the brain, heart and kidneys. MAP is calculated using the formula MAP = (SBP + 2 × DBP) / 3, where SBP is systolic blood pressure and DBP is diastolic blood pressure. This equation is weighted toward diastolic pressure because the heart spends more time in diastole, the resting and filling phase, than in systole, the contracting phase. A normal MAP typically falls around 70-100 mmHg, while a MAP below 65 mmHg is often considered abnormal in critically ill patients and may indicate hypotension, poor organ perfusion or shock risk, warranting urgent clinical review. A MAP above approximately 110 mmHg may also be abnormal and can suggest increased vascular pressure, severe hypertension or significant cardiovascular strain. However, MAP should always be considered alongside other clinical indicators, including heart rate, oxygen saturation, respiratory rate, neurological status and the patient’s overall condition. In an emergency department setting, an abnormal MAP is a clinically significant finding that should prompt thorough patient assessment rather than be dismissed as a recording error.